

Experiment No: 2

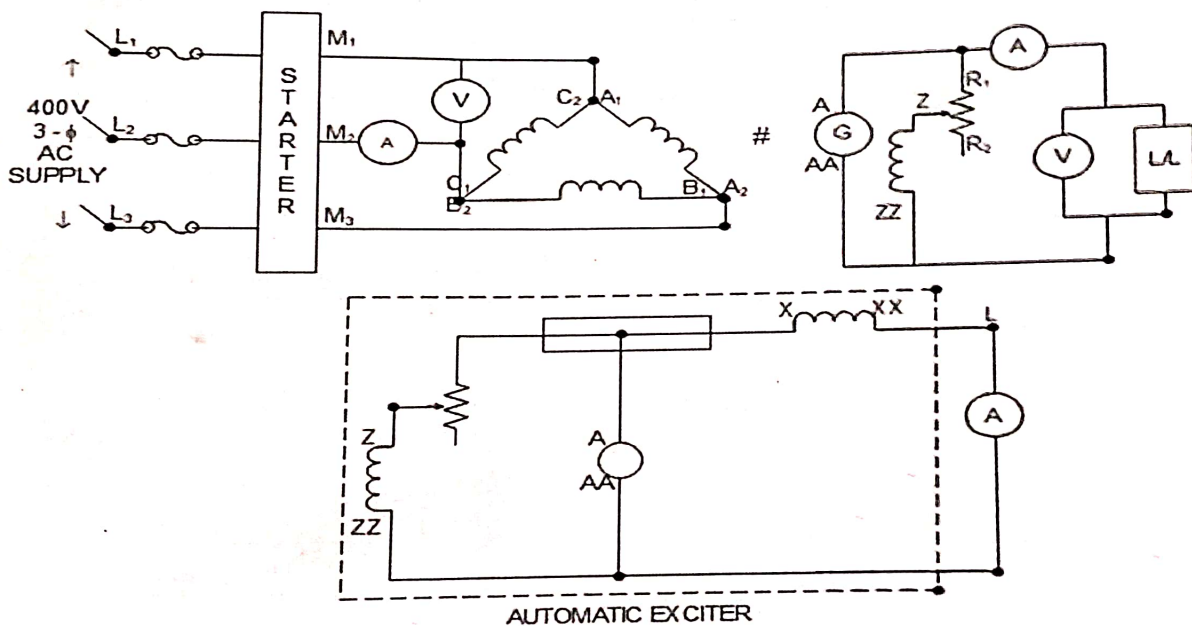
Aim:

To plot V-curves of Synchronous motor on no-load, 25%, 50% & full load

Apparatus Required:

S. NO.	INSTRUMENTS	SPECIFICATIONS	QUANTITY
1.	VOLTMETER (MI)	(0-600V)	1
2.	AMMETER (MC)	(15/30A)	1
3.	AMMETER (DC)	(25/50A)	1
4.	AMMETER (DC)	(10/20A)	1

Circuit Diagram :



Theory:

For a synchronous motor working on a fixed voltage and frequency, the airgap flux is a constant, being contributed by both armature and field currents. The value of the field current at which the airgap flux is totally established by the field is referred to as 'normal' excitation. If the field current is reduced below this value, the armature current is magnetizing and its value is such as to keep the total airgap flux a constant. Similarly if the field current is increased above normal the armature current is demagnetizing and its value will be such as to keep the air gap flux again a constant.

On no load, a variation of field current will thus cause & variation of armature current. On load, the variation of the field current will affect only the reactive component of the armature current, V curve gives a graphical relationship between the armature current and field current of the synchronous motor.

V curve is a plot of the stator current versus field current for different constant loads. The Graph plotted between the armature current (I_a) and field current (I_f) at no load the curve is obtained known as V Curve. Since the shape of these curves is similar to the letter "V", thus they are called the V curve of a synchronous motor.

The power factor of the synchronous motor can be controlled by varying the field current I_f . As we know that the armature current I_a changes with the change in the field current I_f . Let us assume that the motor is running at NO load. If the field current is increased from this small value, the armature current I_a decreases until the armature current becomes minimum. At this minimum point, the motor is operating at a unity power factor. The motor operates at a lagging power factor until it reaches up to this point of operation.

If now, the field current is increased further, the armature current increases and the motor starts operating as a leading power factor. The graph drawn between armature current and field current is

known as the V curve. If this procedure is repeated for various increased loads, a family of curves is obtained.

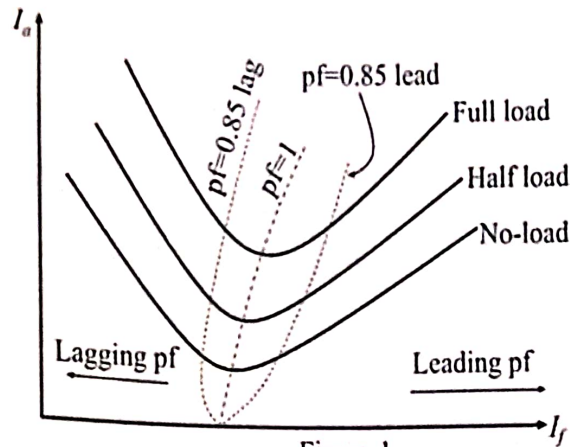


Figure-1
Fig1. V curves of a synchronous motor

The V curves are useful in adjusting the field current. Increasing the field current, If beyond the level for minimum armature current results in leading power factor. Similarly decreasing the field current below the minimum armature current results in a lagging power factor. It is seen that the field current for the unity power factor at full load is more than the field current for unity power factor at no load.

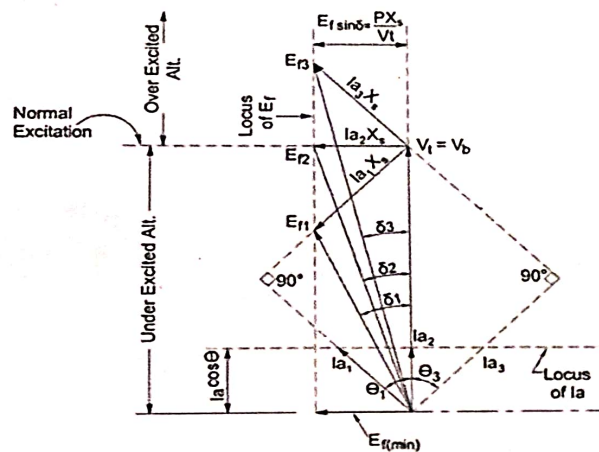


Fig 2. Phasor Diagram

Procedure:

1. Connections are made as shown in diagram.

2. The synchronous motor is started with the help of starter. When the motor attains a speed near about synchronous speed the field excitation is applied automatically and the motor 'pull in'.
3. The excitation of the coupled d.c. generator is adjusted so that It builds up rated voltage.
4. A set of readings of I_f and I_a are taken with the synchronous motor on no load.
5. The synchronous motor is then loaded by loading the d.c. generator sets of readings of i_f and i_a are noted for 25% 50%, 75% and 100% load on the motor. The readings are tabulated for the different load conditions.

Observation table :

S.no.	no load		25% load		50% load	
	Field current (i_f)	Armature Current (i_a)	Field current (i_f)	Armature Current (i_a)	field current(i_a)	armature current (i_f)
1.						
2.						
3.						
4.						
5.						
6.						

Result:

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Precautions:

1. Do not touch any terminal or switch without ensuring that it is dead.
2. Never work alone in the laboratory.
3. Make sure that all the connections are right, before switching on any circuit , faulty connections may cause drawing large power resulting in damage of parts of instruments .
4. The circuit should be de-energized before changing any connections.